Tanay Chitlur

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Summary

Carnegie Mellon computer science student and published machine learning researcher. First author on two peer-reviewed IEEE papers (EMBC, Sensors) applying transformer architectures to biomedical signal synthesis. Two-time ISEF Grand Award winner and 2026 Coca-Cola Scholar.

Education

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Computer Science — *Aug. 2026 – May 2030*

- **Relevant Coursework:** Mathematical Foundations of Computer Science (Discrete Math), Principles of Imperative Computation, Matrices and Linear Transformations, Calculus 1–3.

Westview High School — Portland, OR
Honors Diploma — GPA: 3.98 Unweighted / 4.52 Weighted — *Graduated May 2026*

Research and Publications

A Morphologically-Aware Transformer Network for High-Fidelity V1–V5 ECG Lead Generation — October 2025

- First-authored and presented a peer-reviewed paper at the 24th IEEE Conference on Sensors; indexed in IEEE Xplore (Link).

Transformer-Based Framework with Dynamic Multiscale Attention Mechanisms for 12-Lead ECG Synthesis — July 2025

- First-authored and presented a peer-reviewed paper at the 47th IEEE EMBC, the largest biomedical engineering conference; indexed in IEEE Xplore and PubMed (Link).

IntelliCane: A Deep Learning Cane for Active Obstacle Avoidance & Navigation for the Visually Impaired — May 2025

- Designed a deep learning navigation system for real-time obstacle avoidance; won the AAAI Special Award at Regeneron ISEF (Link).

Validation of Einthoven and Goldberger's Electrocardiographic Lead Systems — May 2024

- Published and presented validation research at the Heart Rhythm Society Annual Meeting (HRS'24) (Link).

Synthesis of a Diagnostic-Caliber 12-Lead Electrocardiogram Using a Novel Multi-resolution Transformer — May 2024

- Built a multi-resolution transformer for 12-lead ECG synthesis; presented at Regeneron ISEF and HRX 2024 (Link).

Early Detection and Synthesis of a 12-Lead ECG Using Density-Based Clustering and GANs — June 2023

- Presented generative modeling research at AI Med 2023, the youngest of 80+ pre- and post-doctoral presenters (Link).

Work Experience

Machine Learning Engineering Intern — June 2024 – July 2026
Hello Alfred.ai — *Austin, TX*

- Built an ML-driven medical history bot for a 24/7 AI health companion using Python, LangChain, OpenAI API (GPT-4), and Pinecone, triaging patient history to flag when clinical care is needed.
- Presented the system at the 2025 Baylor Scott & White Tech4Better Symposium.

Founder — July 2019 – Present
TnT Treats® Baking LLC — *Portland, OR*

- Founded and scaled an artisanal dessert business to $15K+ in lifetime sales (Website).

Team Captain & Software Lead — November 2021 – March 2025
FIRST Tech Challenge — Team 18108, High Voltage — *Portland, OR*

- Wrote autonomous software leveraging OpenCV pipelines and RoadRunner for PID-based motion profiling. Led hardware, software, and outreach initiatives, culminating in the Control Award at the Oregon State Championship.

Awards

Coca-Cola Scholar, 2026 — $20,000 merit-based national scholarship; 1 of 150 selected from 110,000+ applicants
ISEF Grand Award, 2nd Place — Robotics and Intelligent Machines, Regeneron ISEF, 2024
ISEF Grand Award, 3rd Place & AAAI Special Award — Robotics and Intelligent Machines, Regeneron ISEF, 2025

Technical Skills

Languages: Python, C++, Swift, Java, JavaScript, HTML, CUDA
ML & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, XGBoost
Data & Tools: Pandas, NumPy, SciPy, Matplotlib, Seaborn, h5py, React, Linux, AWS, Fusion 360
Developer Tools: Git, Cursor, Claude Code, GPT Codex